

<u>Steps</u>	<u>Reason</u>
2. $\forall x (DM(x) \Rightarrow Al(x))$	Premise
3. $DM(Melissa) \wedge DM(Aaron)$ $\wedge DM(Ralph) \wedge DM(Veneesha)$	Simplification on 1
4. $DM(Veeshawn) \wedge DM(Melissa)$ $\wedge DM(Aaron) \wedge DM(Ralph) \wedge$ $DM(Veneesha)$	Commutative law on 1
5. $DM(Veeshawn)$	Simplification on 4
6. $DM(Melissa) \wedge DM(Aaron)$ $\wedge DM(Ralph)$	Simplification on 3
7. $DM(Veneesha) \wedge DM(Melissa)$ $\wedge DM(Aaron) \wedge DM(Ralph)$	Commutative law on 3
8. $DM(Veneesha)$	Simplification on 7
9. $DM(Melissa) \wedge DM(Aaron)$	Simplification on 6
10. $DM(Ralph) \wedge DM(Melissa) \wedge$ $DM(Aaron)$	Commutative law on 6
11. $DM(Ralph)$	Simplification on 10
12. $DM(Melissa)$	Simplification on 9
13. $DM(Aaron) \wedge DM(Melissa)$	Commutative law on 9
14. $DM(Aaron)$	Simplification on 13
15. $DM(Melissa) \Rightarrow Al(Melissa)$	Universal instantiation on 2
16. $Al(Melissa)$	Modus ponens on 12, 15
17. $DM(Aaron) \Rightarrow Al(Aaron)$	Universal instantiation on 2
18. $Al(Aaron)$	Modus ponens on 14, 17
19. $DM(Ralph) \Rightarrow Al(Ralph)$	Universal instantiation on 2
20. $Al(Ralph)$	Modus ponens on 11, 19
21. $DM(Veneesha) \Rightarrow Al(Veneesha)$	Universal instantiation on 2